Amanda Fonseca Rodrigues

Software Developer | Frontend & Full-Stack

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Professional Summary

Computer Science undergraduate with experience in software development, web applications, frontend, and full-stack contributions. Focuses primarily on React, TypeScript, JavaScript, Node.js, and REST API integration. Experienced in requirements analysis, UI development, feature organization, team leadership, and applied research. Throughout academic training, participated in institutional and academic projects related to software engineering, digital health, artificial intelligence, Internet of Things, and Information Security.

Professional Experience

Instituto Atlântico *2026 – Present*

Innovation Fellow – Software Development

- Work on a strategic innovation and artificial intelligence project, contributing to the development, evolution, and validation of technological solutions targeted at industrial sector challenges.
- Participate in software development activities, requirements analysis, feature structuring, and acceptance criteria definition.
- Contribute to workflow and user journey modeling, connecting user needs, project goals, and technical implementation.
- Collaborate with multidisciplinary teams across different stages of the solution development lifecycle.

PET-Saúde Digital – University of Fortaleza (Unifor) *2026 – Present*

PET-Saúde Digital Fellow

- Participate in the development of the Electronic Call Panel, a solution created to support care organization and optimize patient flow in public healthcare.
- Collaborate within an interdisciplinary team involving technology, academia, and healthcare professionals from the Brazilian Unified Health System (SUS).
- Contribute to application updates based on real-world clinical workflow requirements.

COSMOSTECH *2025 – 2026*

Frontend Developer & Full-Stack Contributor – Volunteer

- Participated in the frontend development of a web application using React and TypeScript.
- Implemented user interfaces, state management, and REST API integrations.
- Made targeted backend contributions and collaborated on development processes using Git, branching strategies, and deployment.

VORTEX Laboratory – PRISMA IAT Project – Unifor *2025*

Development Team Lead – Volunteer

- Led a team of three developers building PRISMA IAT, a psychometric software designed to execute implicit association tests in research and educational contexts.
- Organized priorities, assigned tasks, and tracked deliverables, serving as the main communication point between developers, faculty, and stakeholders.
- Contributed to the full-stack development of the application using React and TypeScript.
- Assisted in backend organization following Model, Repository, and Service separation of concerns.

Federal Institute of Ceará (IFCE) *2021*

Research Fellow – Machine Learning and Internet of Things

- Researched Machine Learning techniques applied to intrusion detection in Internet of Things (IoT) environments.
- Implemented and evaluated models using Python and Scikit-learn, analyzing performance metrics and experimental results.
- Developed experience in scientific research, programming, data analysis, and Information Security.

Selected Projects

VetCare – Veterinary Management System GitHub

- Developed a full-stack application using Node.js, Sequelize, MySQL, React, and TypeScript.
- Implemented appointment scheduling, customer management, and financial tracking features, integrating frontend, backend, and database.

Stock Portfolio Management System Backend | Frontend

- Developed a full-stack application using Clojure for the backend and ClojureScript with Reagent for the frontend.
- Integrated with the Brapi API and implemented portfolio management, transaction history, financial indicators, and interactive dashboards.

CinemaGOF – Microservices with Java and Spring Boot GitHub

- Developed a distributed ticket management system using Java and Spring Boot, structured into modular services communicating via REST APIs.

Technical Skills

Frontend: React.js, TypeScript, JavaScript, Bootstrap, React Native, Tailwind CSS, component architecture, state management, REST API integration.

Backend: Node.js, Express.js, Java, Spring Boot, Clojure, REST APIs.

Databases: PostgreSQL, MySQL, MongoDB, Sequelize.

Python & Data: Python, Pandas, NumPy, Scikit-learn, Machine Learning fundamentals.

Software Engineering: Requirements analysis, acceptance criteria, software architecture, microservices, technical documentation, agile methodologies.

Tools & DevOps: Git, GitHub, Docker, Vercel, CI/CD.

Education

University of Fortaleza (Unifor) *2023 – Present*
Bachelor of Science in Computer Science

Federal Institute of Ceará (IFCE) *2020 – 2023*
Technical Degree in Computer Networks

Publications

- CASTELO, J. S. F.; RODRIGUES, A. F.; CASTRO, I. C. C.; MEIO FILHO, J. I. B.; ALMEIDA, L. B. C. *From theory to practice: an experience report on the development of the PRISMA psychometric software*. 2025.
- RODRIGUES, A. F.; CARLA M. S.; LUMA B. C. A. *Collaborative management of university sanitation: an extension project for social and technological engagement*. 2024.

Languages

Portuguese: Native
English: Intermediate (proficient in technical documentation reading)